

HDOC Day 22

Menu-Driven C Program Using switch

Question

Write a menu-driven C program that accepts two positive integers, m and n, where $m \leq n$. Display a menu and use a switch statement to perform the operation selected by the user:

1. Print the multiplication tables of m and n side by side, using multipliers from 1 to 10.
2. Display a multiplication grid containing the products of integers from 1 to m and integers from 1 to n.
3. For each integer from m to n, print that integer followed by the next five consecutive integers.
4. Exit the program.

Continue displaying the menu until the user selects Exit. Display an appropriate error message for an invalid menu choice.

Note: Prepare the algorithm and test-case table. Then write, compile and execute the C program, and test it against the prepared test cases.

Algorithm

1. Start.
2. Read two positive integers m and n.
3. If $m > n$ or either value is not positive, display an error message and ask for valid values.
4. Display the four menu options.
5. Read the user choice.
6. Use a switch statement to process the choice.
7. For choice 1, use a loop from 1 to 10 and print the multiplication tables of m and n side by side.
8. For choice 2, use nested loops to print a multiplication grid with rows 1 to m and columns 1 to n.
9. For choice 3, for every integer i from m to n, print i followed by i+1, i+2, i+3, i+4 and i+5.
10. For choice 4, terminate the program.
11. For any other choice, display an invalid-choice message.
12. Repeat the menu until choice 4 is selected.
13. Stop.

C Program

```
#include <stdio.h>
```

```

int main() {
    int m, n, choice, i, j, k;

    printf("Enter two positive integers m and n (m <= n): ");
    scanf("%d %d", &m, &n);

    if (m <= 0 || n <= 0 || m > n) {
        printf("Invalid input. Enter positive integers with m <= n.\n");
        return 0;
    }

    do {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Multiplication Tables of %d and %d (m=%d, n=%d)\n", m, n, m, n);
                for (i = 1; i <= 10; i++)
                    printf("%d x %d = %-4d  %d x %d = %d\n", m, i, m*i, n, i, n*i);
                break;

            case 2:
                printf("Multiplication Grid (m=%d, n=%d)\n", m, n);
                printf(" ");
                for (j = 1; j <= n; j++) printf("%4d", j);
                printf("\n ");
                for (j = 1; j <= n; j++) printf("----");
                printf("\n");
                for (i = 1; i <= m; i++) {
                    printf("%2d |", i);
                    for (j = 1; j <= n; j++) printf("%4d", i*j);
                    printf("\n");
                }
                break;

            case 3:
                printf("Consecutive Integers from %d to %d (m=%d, n=%d)\n", m, n, m, n);
                for (i = m; i <= n; i++) {
                    printf("%d:", i);
                    for (k = 0; k <= 5; k++) printf(" %d", i+k);
                    printf("\n");
                }
                break;

            case 4:
                printf("Exiting program.\n");
                break;

            default:
                printf("Invalid menu choice. Please select 1 to 4.\n");
        }
    } while (choice != 4);

    return 0;
}

```

Test-Case Table

Test Case	Input / Choice	Expected Output	Actual Output	Result
1	m=3, n=5; choice=1	Tables of 3 and 5 from x1 to x10	Tables displayed correctly	Pass
2	m=3, n=5; choice=2	3 x 5 multiplication grid	Grid displayed correctly	Pass
3	m=3, n=5; choice=3	Rows for 3, 4 and 5 with next five integers	Consecutive integers displayed correctly	Pass
4	m=3, n=5; invalid choice	Invalid menu choice message	Error message displayed	Pass
5	m=3, n=5; choice=4	Program exits	Program exits correctly	Pass

Program Output Screenshots

Dear Student,

The following are the **HDOC Day-22** activities for Batches 15 & 16:

Write a menu-driven C program that accepts two positive integers, m and n, where $m \leq n$. Display the following menu and use a switch statement to perform the operation selected by the user:

- 1. Print the multiplication tables of m and n side by side, using multipliers from 1 to 10.*
- 2. Display a multiplication grid containing the products of integers from 1 to m and integers from 1 to n.*
- 3. For each integer from m to n, print that integer followed by the next five consecutive integers.*
- 4. Exit the program.*

Continue displaying the menu until the user selects the Exit option. Display an appropriate error message if the user enters an invalid menu choice.

Note: *Prepare the algorithm and test-case table. Then write, compile and execute the C program, and test it against the prepared test cases.*

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 1

Multiplication Tables of 3 and 5 (m=3, n=5)

3 x 1 = 3	5 x 1 = 5
3 x 2 = 6	5 x 2 = 10
3 x 3 = 9	5 x 3 = 15
3 x 4 = 12	5 x 4 = 20
3 x 5 = 15	5 x 5 = 25
3 x 6 = 18	5 x 6 = 30
3 x 7 = 21	5 x 7 = 35
3 x 8 = 24	5 x 8 = 40

Output: Menu and multiplication tables (Choice 1)

3 x 9 = 27 5 x 9 = 45
3 x 10 = 30 5 x 10 = 50

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 2

Multiplication Grid (m=3, n=5)

```
    1  2  3  4  5
-----
1 |  1  2  3  4  5
2 |  2  4  6  8 10
3 |  3  6  9 12 15
```

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 3

Consecutive Integers from 3 to 5 (m=3, n=5)

3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

Output: Multiplication grid and consecutive integers (Choices 2 and 3)

Result

The menu-driven C program was written, compiled and executed successfully. It correctly performs the multiplication-table, multiplication-grid and consecutive-integer operations using a switch statement and repeatedly displays the menu until Exit is selected.

Program Execution and Output Screenshots

```
(kali㉿kali)-[~/Desktop/apple]
$ gcc hdoc22.c -o hdoc22

(kali㉿kali)-[~/Desktop/apple]
$
```

Compilation of the C program

```
(kali㉿kali)-[~/Desktop/apple]
$ ./hdoc22
Enter two positive integers m and n ( $m \leq n$ ): 3
5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1
Multiplication Tables of 3 and 5 (m=3, n=5)
3 x 1 = 3      5 x 1 = 5
3 x 2 = 6      5 x 2 = 10
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50
```

Execution with input m = 3, n = 5 and Menu Option 1: Multiplication Tables

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 2

Multiplication Grid (m=3, n=5)

	1	2	3	4	5
1	1	2	3	4	5
2	2	4	6	8	10
3	3	6	9	12	15

Menu Option 2: Multiplication Grid

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 3

Consecutive Integers from 3 to 5 (m=3, n=5)

3: 3 4 5 6 7 8

4: 4 5 6 7 8 9

5: 5 6 7 8 9 10

Menu Option 3: Consecutive Integers

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 7

Invalid menu choice. Please select 1 to 4.

Invalid menu choice handling

```
1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4
Exiting program.
```

```
(kali@kali)-[~/Desktop/apple]
$
```

Menu Option 4: Exit

Final Result

The menu-driven C program was compiled and executed successfully. All menu options produced the expected output for $m = 3$ and $n = 5$, invalid menu input was handled correctly, and the program exited successfully when option 4 was selected.